4.1.3 Trace amounts of air will condense or solidify in an environment with liquid hydrogen possibly resulting in an unstable and explosive mixture. Precautions should be taken to assure that the possibility of condensed air is accounted within properly secured hazard areas.
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK
MSC 108/20/Add.1
Annex 20, page 10
I:\MSC\108\MSC 108-20-Add.1.docx
4.2 Low temperature hazard
4.2.1 Selection of appropriate material
4.2.1.1 Tables 6.3 and 6.4 in the Code prescribe material selection for piping or cargo tanks whose design temperature is -165°C or higher. According to Note 2 of table 6.3 and Note 3 of table 6.4 of the Code, the requirements for materials whose design temperatures are lower than -165°C should be specially agreed with the Administration. In this regard, the publication
---
SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
than -165°C should be specially agreed with the Administration. In this regard, the publication
by AIAA2) introduces some appropriate materials corresponding to the design temperature and the Administration should take into account such references for the material selection.
4.2.1.2 Although paragraph 4.19.3 in the Code requires testing of materials used for thermal insulation for various properties adequate for the intended service temperature, the minimum test temperature is -196°C. The requirements in the Code do not refer to the normal boiling point of hydrogen, being -253°C. In case of carriage of liquefied hydrogen, special requirements should be provided to consider the lower design temperature.
4.2.2 Measures for condensed air
4.2.2.1 In the case of nitrogen whose normal boiling point is -196°C, for which air condensation and oxygen enrichment are concerns, the following special requirement has already been included in paragraph 17.17 in the Code:

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
condensation and oxygen enrichment are concerns, the following special requirement has
already been included in paragraph 17.17 in the Code:
"Material of construction and ancillary equipment such as insulation shall be resistant to the effect of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system. Due consideration shall be given to ventilation in such areas where condensation might occur to avoid the stratification of oxygen-enriched atmosphere."
A similar special requirement is applicable to hydrogen.
4.2.2.2 A vent may be blocked by accumulation of ice formed from moisture in the air, which may result in excessive pressure leading to rupture of the vent and relevant piping (see paragraph 4.2.4).
4.2.3 Removal of impure substances condensed
The removal of impure substances, such as those contained in condensate in pipes, should
---
SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).p

df
4.2.3 Removal of impure substances condensed
The removal of impure substances, such as those contained in condensate in pipes, should be separately considered. Installation of filters can be an appropriate measure and should be stipulated as a special requirement.
4.2.4 Prevention of blockage due to formation of water or ice
Pressure relief systems may become blocked due to formation of water or ice, depending on the temperature and humidity of air, resulting from the low temperature of the cargo and its vapour (see paragraph 4.2.2) . Appropriate means should be provided to prevent such phenomena.
4.3 Hydrogen embrittlement
4.3.1 Selection of appropriate materials should be required to prevent failures owing to hydrogen embrittlement. The publication by AIAA 2) introduces some appropriate materials resistant to hydrogen embrittlement, and concludes that aluminium is the material least affected.
RESOLUTION MSC.565(108) (adopted on 24 May 2024)

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
resistant to hydrogen embrittlement, and concludes that aluminium is the material least affected.
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF
LIQUEFIED HYDROGEN IN BULK
MSC 108/20/Add.1
Annex 20, page 11
I:\MSC\108\MSC 108-20-Add.1.docx
4.3.2 International or national standards should be followed for the selection of materials for the design of liquefied and gaseous hydrogen installations in a marine environment.
4.4 Susceptibility to leakage
4.4.1 Prevention of leakage from pipes
To mitigate undetected accumulation of hydrogen in a confined space, effective measures should be employed to reduce the possibility of leakage of hydrogen, taking its leakage characteristics into account. Effective measures can be double tube structures, or fixed hydrogen leak detectors in areas assessed as being highly hazardous with regard to hydrogen
---
SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
hydrogen leak detectors in areas assessed as being highly hazardous with regard to hydrogen leakage. Hydrogen leakage through welds, joints and seals is an important consideration for the design of hydrogen systems and an important operational issue.
4.4.2 Implementation of effective tightness test
4.4.2.1 Tightness tests for cargo tanks and cargo pipes/valves are required by paragraphs 4.20.3.2, 5.13.1 and 5.13.2.3 in the Code respectively. Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the medium for tightness tests, instead of air, because hydrogen is highly susceptible to leakage.
4.4.2.2 For a hydrogen installation, the pipework should be pressure -tested at its design pressure. Consideration should be given to using oxygen-free nitrogen with a small molecule tracer gas, such as helium as the test medium and an electronic leak detec tor for identifying leaks.

4.4.3 Confirmation of appropriate operating procedure

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
leaks.
4.4.3 Confirmation of appropriate operating procedure
Instructions/manuals containing the operating procedures for the prevention of leakage during transport, methods for early detection in case of leakage, and appropriate measures after such events, should be provided. For this, paragraph 18.3 of the Code requires that the information shall be on board and available to all concerned, giving the necessary data for the safe carriage of cargo. In detail, the Code requires such information on action to be taken in the event of spills or leak, countermeasures agai nst accidental personal contact, procedures for cargo transfer, and emergency procedures to be on board. With regard to the manuals on procedures for liquefied hydrogen during carriage and transfer operations, the requirements in the Code are applicable and no special requirement is necessary.
4.5 Low density and high diffusivity

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
are applicable and no special requirement is necessary.
4.5 Low density and high diffusivity
Though low density and high diffusivity of hydrogen may reduce the possibility of formation of a flammable atmosphere in open spaces, adequate ventilation is necessary for enclosed spaces in cargo areas where formation of hydrogen -oxygen/air mixture may occur. Paragraph 12.2 of the Code requires fixed ventilation systems or portable mechanical ventilation for such enclosed spaces. These requirements in the Code are applicable to liquefied hydrogen carriers and no special requirement is necessary in this regard.
4.6 Ignitability
4.6.1 The Code requires electrical bonds of the piping and the cargo tanks in paragraph 5.7.4, exclusion of all sources of ignition in paragraph 11.1.2, electrical installations to minimize the risk of fire and explosion from flammable products in paragraph 10.2.1 and so on, in order to prevent ignition of flammable cargoes.

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
on, in order to prevent ignition of flammable cargoes.

4.6.2 The Code requires compliance with the relevant standards issued by the International Electrotechnical Commission (IEC) and the IEC standards specify the details of such safety measures depending on the respective properties of flammable gases including hydrogen. No special requirement is necessary with regard to ignitability of hydrogen.*
4.7 Fire hazard
4.7.1 Safety of personnel in case of fire

To avoid the effects of flame and UV radiation produced by a hydrogen fire, it is effective to use fire-fighter's outfits and protective equipment. The Code already requires fire -fighter's outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in
---
SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
outfits for ships carrying flammable products in paragraph 11.6.1 and safety equipment in paragraph 14.3. This issue should be considered as the matter of cargo information required by paragraph 18.3 of the Code. Due consideration should be given to the invisible nature of hydrogen fire.

4.7.2 Compatibility of fire-extinguishing systems

Dry chemical powder fire -extinguishing or carbon dioxide fire -extinguishing systems are considered to be effective in case of hydrogen fire and such fire -extinguishing systems are already required by paragraphs 11.4 and 11.5 of the Code. Special requiremen ts for installation of other types of fire -extinguishing systems are considered unnecessary, except with regard to the increased of amount of carbon dioxide required, as mentioned in the next paragraph in this document.

4.7.3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems

4.7.3.1 Paragraph 11.5.1 of the Code requires as follows:

SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
4.7.3 Increase of the amount of gas for carbon dioxide fire-extinguishing systems

4.7.3.1 Paragraph 11.5.1 of the Code requires as follows:

"Enclosed spaces meeting the criteria of cargo machinery spaces in 1.2.10, and the cargo motor room within the cargo area of any ship, shall be provided with a fixed fire-extinguishing system complying with the provisions of the FSS Code and taking into account the necessary concentrations/application rate required for extinguishing gas fires."

4.7.3.2 Chapter 5 of the FSS Code, i.e. Fixed gas fire -extinguishing systems, requires that the quantity of carbon dioxide for cargo spaces, unless otherwise provided, shall be sufficient to give a minimum volume of free gas equal to 30% of the gross volume of the largest cargo space to be protected in the ship, in paragraph 2.2.1.1.

4.7.3.3 On the other hand, NFPA 123) requires that the design quantity of carbon dioxide for
---
SOURCE: MSC.565(108).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
4.7.3.3 On the other hand, NFPA 123) requires that the design quantity of carbon dioxide for
hydrogen fire should be 75% or more of the gross volume of the protected space. The special